

Refrigerator and Heat Pump

- ❖ To transfer heat from a low-temperature region to a high-temperature, one requires special devices called Refrigerators.
- ❖ Refrigeration keeps a space temperature lower than that of its surroundings by continuously pumping the heat from space to surroundings.
- ❖ Refrigerators and heat pumps are essentially the same devices; they differ in their objectives only.
- ❖ The objective of a refrigerator is to remove heat (Q_L) from the cold medium; the objective of a heat pump is to supply heat (Q_H) to a warm medium

NEED FOR INSULATION

Since heat always travels from a region of higher temperature to a region of lower temperature, there is always a continuous flow of heat into the refrigerated region from the warmer surroundings.

To limit the flow of heat in to the refrigerated region to some practical minimum, it is necessary to isolate the region from its surroundings with a good heat insulating material.

Refrigeration Capacity

- ❑ Capacity of refrigerating machines are expressed by their cooling capacity. The commonly used unit for expressing the capacity of a refrigerating machine is Ton of refrigeration.
- ❑ One ton of refrigeration is defined as the quantity of heat extracted (refrigerating effect) to freeze one short ton of water from at 0°C into ice in 24 hours.
- ❑ 1 Short ton = 2000 pounds = 907 kg (in US a short ton is called as 'ton')
- ❑ Latent heat of ice = 335 kJ/kg i.e., 335 kJ of heat should be extracted from one kg of water at 0°C to convert it into ice.
- ❑ So one Ton of refrigeration = $335 \times 907 \text{ kJ} / 24 \text{ hrs} = 335 \times 907 / (24 \times 60 \times 60) \text{ kJ/s}$
= 3.5 kW.

Performance of refrigeration

Coefficient of Performance (COP)

- Carnot COP of Refrigerator = $T_L / T_H - T_L$
- Carnot COP of Heat Pump = $T_H / T_H - T_L$
- COP of Refrigerator is a ratio of heat absorbed by refrigerant while passing through the evaporator to work input required to compressor the refrigerant in compressor.

$$\text{COP} = (\text{Net refrigerant effect} / \text{Work done})$$

- COP of Heat Pump is a ratio of heat rejected by refrigerant while passing through the condenser to work input required to compressor the refrigerant in compressor.

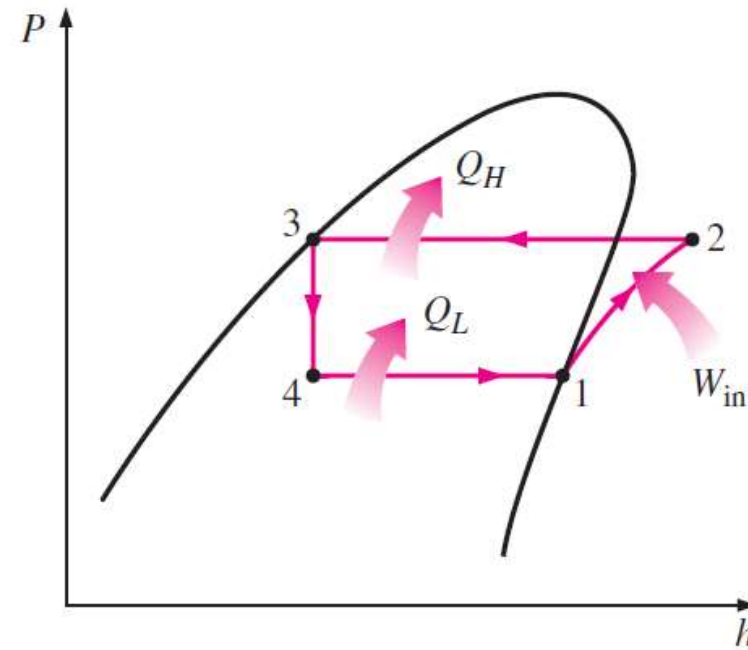
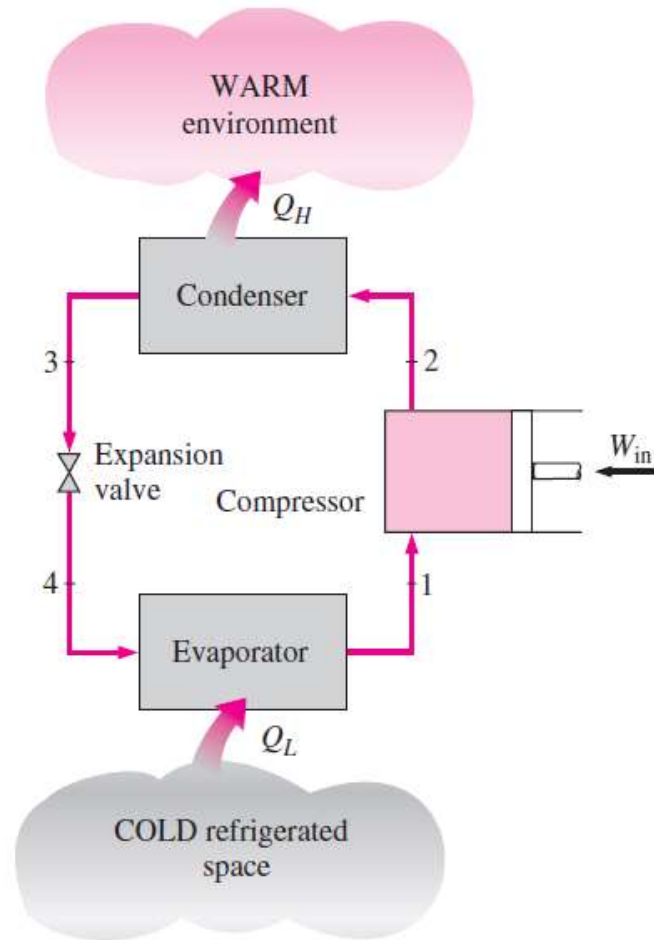
$$\text{COP} = (\text{Net heating effect} / \text{Work done})$$

- HERBERT HOOVER WAS THE FIRST PRESIDENT TO ENJOY AIR CONDITIONING. HE SPENT \$30,000 TO INSTALL THE SYSTEM IN THE OVAL OFFICE.
- IF YOU WANTED TO BUY A BASIC AC UNIT IN THE 1940'S, IT WOULD COST YOU APPROXIMATELY \$350. TRANSLATING THAT INTO TODAY'S PRICE BASED ON INFLATION, YOU WOULD BE PAYING ALMOST \$3,500
- THE FIRST CAR WITH OPTIONAL AIR CONDITIONING WAS INTRODUCED IN 1939 BY THE AUTOMOBILE MANUFACTURER PACKARD. IT WASN'T VERY POPULAR DUE TO ITS HIGH COST AND THE FACT THAT THE EVAPORATOR AND BLOWER SYSTEM TOOK UP HALF THE TRUNK SPACE .

Applications of Refrigeration

- Shopping malls food storage
- Ice making
- Food processing
- Sea food processing/preservation
- Cold storages
- Refrigerated transportation
- Hospital / Health Sciences
- Household/ domestic use
- Cryogenics/ Liquefied gases
- Pharmaceutical industry
- Large industrial plants
- Air-Conditioning for Hot and wet climate

VAPOR COMPRESSION REFRIGERATION SYSTEM



Schematic and p-h diagrams

Components and process in VCR system

Compressor

- ❑ The compressor receives a low pressure and low temperature vapour refrigerant from the evaporator and compresses it to a high pressure to increase its saturation temperature , so that the refrigerant vapor rejects its heat to air or water at atmospheric temperature.

Condenser

- The high pressure and high temperature vapour from compressor enters into the condenser where it converted into liquid by losing its latent heat.
- Commonly, water or air is used as condensing medium in condenser.

Components and process in VCR system

Expansion Device

- ❑ It is used to reduce the pressure of the refrigerant liquid from condenser to the evaporator, so that its saturation temperature reduces to low value to absorb heat from low-temperature space, where evaporator is located.

Evaporator

- In evaporator, the low-pressure liquid refrigerant absorbs the latent heat of the substance placed in its surroundings and get converted into vapour. Then this low-pressure vapour enters into the compressor.

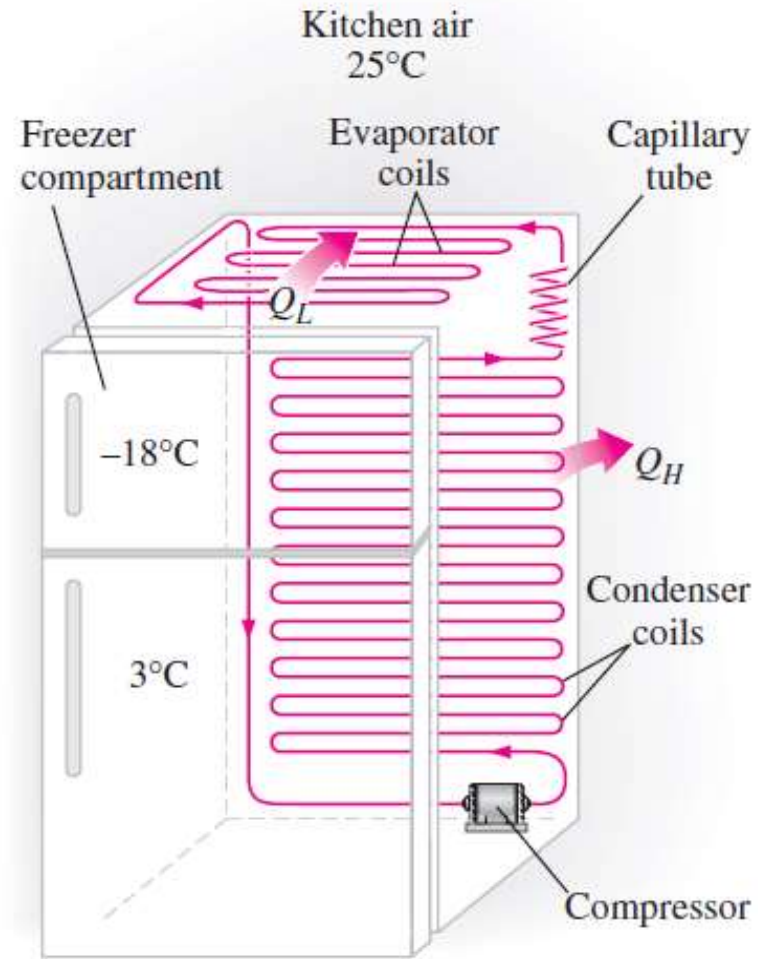
Vapour Compression Refrigeration system

- The vapour-compression refrigeration system is widely used in refrigeration applications like refrigerator, water cooler, cold storage and air conditioner.
- The refrigerant vapour is compressed to a higher pressure and higher temperature as superheated vapour in the compressor.
- The power supply is given to the motor to drive the compressor, and the compressor is a more power utilization unit.
- The condenser receives the compressed refrigerant vapour. The latent heat from the refrigerant vapour is carried away by blowing ambient air over the condenser using a fan. Hence, the refrigerant vapour is condensed into high pressure liquid.

Vapour Compression Refrigeration system

- Liquid refrigerant enters the expansion device which can be a capillary tube or an expansion valve where refrigerant is throttled to low pressure, low temperature wet vapour.
- After expansion, refrigerant enters the evaporator where the refrigeration effect is produced.
- Evaporator section cools the substances stored in it surrounding by absorbing their latent heat of vaporization. In Air-conditioning system, air to be chilled is blown across the evaporator .
- The refrigerant comes out of the evaporator will be dry saturated or slightly superheated state. This saturated or superheated vapour enters the compressor and then cycle repeats.

DOMESTIC REFRIGERATOR



Domestic Refrigeration system

- Domestic refrigeration systems belong to the vapour compression type. The evaporator is the coldest portion and is located in the freezer compartment.
- The condenser tubes are stored in the back of the refrigerator cabinet. The surrounding air condenses the refrigerant vapour, which rises above due to natural convection.
- The evaporator is situated outside the freezer section in a no-frost refrigerator. A fan makes cold air flow into the freezer compartment.
- There are multiple compartments with progressively higher temperatures below freezer sections. There are separate doors for freezer and refrigerator compartment. The bottom most compartment is for vegetables that do not require a very low temperature.

REFRIGERANTS

Working fluid in the Refrigerators / Air conditioners

Inorganic Refrigerants– NH₃, water, CO₂

Organic refrigerants – Halogenated Hydrocarbon

R12 – CCl₂F₂

R22 – CHClF₂

R134 – C₂H₂F₄, etc.

Required Properties:

Low Boiling point, Low freezing point,

High value of latent heat (h_{fg}), high thermal conductivity,

Low Viscosity

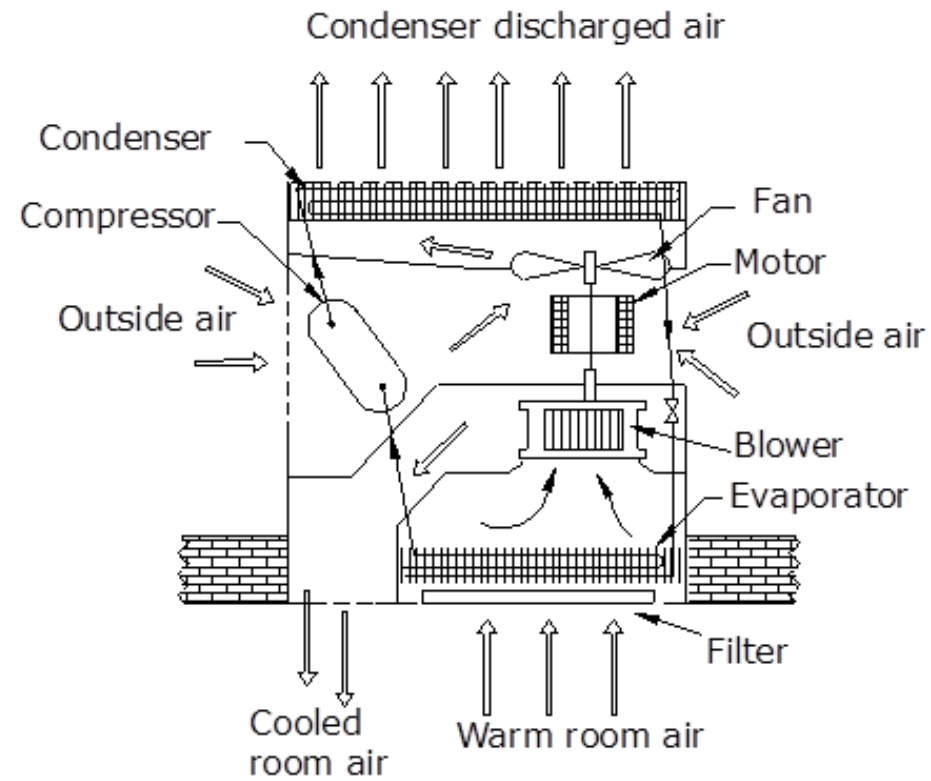
Non-flammable

Non-toxic

Non-corrosive

Low cost, etc.

WINDOW AIR CONDITIONER



WINDOW AIR CONDITIONER

